Sinong Wang

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Research Direction: Deep Learning Algorithm, Digital Image Processing

EDUCATION

2020.9-2023.7(expected)	**M.S.**, Software Engineering, **Northeastern University (NEU)**, Shenyang, China Supervisor: Wei Zhang Overall GPA: **3.6/4.0 (Ranking 3/27)** **Main courses**: Applied mathematical statistics, Advanced Artificial Intelligence, Advanced Software Engineering
2021.1-2022.7(expected)	**M.S.**, Computer Science, **The University of Texas at Arlington (UTA)**, Texas, USA Overall GPA: **4.0/4.0 (Ranking 1/27)** **Main courses**: Machine Learning, Artificial Intelligence, Algorithm, DBMS Models and Implementation, Data Mining, Agile Development, Advanced DBMS
2016.9-2020.7	**B.S.**, Non-woven Material and Engineering, **DongHua University (DHU)**, Shanghai, China

AWARDS & CERTIFICATION

2021-2022	First Class Scholarship in NEU
2020-2021	Second Class Scholarship in NEU
2020-2021	Deep Learning Specialization - Coursera
2017-2018	Scholarship for Outstanding work in Society in DHU

RESEARCH EXPERIRENCE

♠ **2021.04-Present** *Supervisor: Wei Zhang*

DeepLearning-based Super Resolution Algorithm via Feature Adaptive Restore

- **Research Purpose**: Solve the single image super-resolution algorithm based on deep learning from the following two aspects:
  - Adopt **Transformer Encoder** to abstract global features and apply it to whole image patches.
  - Add new constraint to shrink hypothesis space to find a more reasonable convergence position.

- **Research Achievements**:
  a. **Designed a new model** which the performance on test set is in the first echelon, and the details of the model need to be optimized.
  b. **Coded a deep learning training and testing system** based on PyTorch.
  c. Summarized the two main directions to solve super resolution problem.
  d. Familiar with low-level visual tasks, **structured learning program**, and basic model of CNN, Transformer, GAN, etc.

♠ **2021.10-Present** *Supervisor: Wei Zhang*

Deep Learning based Adaptive Colour Channel Real-world Image Denoising

- **Research Purpose**:
  - Use the imaging principle and **basic features of digital image**, more information is extracted through deep learning model to help image restoration.
- **Research Achievements**:
  a. Designed a new model by **combining traditional image processing algorithm and deep learning model**.
  b. Made a **new real-world data set** which contain real-world ground truth through control the different ISO and combine noisy images.
  c. Designed a new Loss Function which consider the interaction of different channels.
  d. Deepened the understanding of deep learning and image processing.

♠ **2021.08-2021.10** *Supervisor: Khalili Barham*

Agile Development Project: Personal Health Monitoring System (PHMS)

- **Research Purpose**:
  - Simulated Agile development of a complete system using Scrum methodBuild a personal health monitoring system
- **Research Achievements**:
  a. Built a complete system through MySQL, Redis, Flask, html, Javascript.
  b. Improved the ability of cooperation and communication.
  c. Won the first place in all the projects.

RELATED EXPERIENCE

Red Packet Delivery Project, ByteDance Back-end Boot Camp 2021.10-2021.12

- Use go language to complete **highly concurrent back-end** code, including token bucket, message queue, etc.

PingAnHaoXue, Online education for teenagers 2020.06-2021.03

- Python teacher(part-time)

SKILLS

- Programming languages: good command of **Python** and **C/C++** programming languages.
- Development Environment: Familiar with **Linux** development environment.
- Platform: **PyTorch**, **OpenCV**
- English level: CET-6 and preparing TOEFL & GRE.